

Jaskaran Sarkaria

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Education

University of Manitoba

September 2019 - November 2025

Bachelor of Science in Computer Engineering – Co-op/IIP Option

Engineering Work Experience

Database Engineer

June 2025 - Current

Vosyn

Remote

- Improved workforce tracking accuracy by ~40% through development of a React + Node.js clock-in system using SDLC principles hosted on Google Cloud Platform, automating attendance, holiday, and leave management for 800+ employees.
- Achieved 5x faster API response handling through deployment of a Virtual Large Language Model (vLLM)-based server using Docker for an internal AI chatbot using (OpenAI, Gemini 2.5 Pro) through RAG, enabling scalable document retrieval, backend integration, and automated workflows via n8n.
- Optimized backend data reliability and scalability by designing and improving PostgreSQL database schemas on Google Cloud Platform (GCP), implementing indexing, normalization, and IAM-based access control aligned with cloud architecture best practices.
- Improved backend security and code quality by reviewing the PostgreSQL and Node.js codebase for injection vulnerabilities and enforcing a modular, coherent architecture.

Computer Engineering Intern – Safety & Construction Services

January 2023 – December 2023

TC Energy

Calgary, AB

- Enhanced project visibility for 1,000+ employees through development of modular Power Apps and Power BI dashboards using JavaScript, Power Fx, and REST API calls, applying structured data-flow abstraction and software engineering best practices on the Microsoft Power Platform.
- Improved data access for an executive team of 30+ through configuring and maintaining departmental web pages using JavaScript and HTML, working directly with the Director to sustain project health and operations after major team layoffs.
- Developed an automated data-extraction tool using Python and Power Automate, processing data from 300+ sources in collaboration with cross-functional teams and senior leadership improving reporting efficiency.

Engineering Project Experience

Drone-Based Pothole Detection System

September 2024 – April 2025

University of Manitoba

- Achieved 91% detection accuracy by developing an AI-driven computer-vision model on Raspberry Pi Zero 2W using YOLOv11, PyTorch, OpenCV, and Roboflow for real-time edge inference with a 4K camera and GPS.
- Improved model robustness and precision through data preprocessing, augmentation, and hyperparameter tuning, integrating attention mechanisms (C2PSA, C3k2) for small-object detection.
- Deployed the trained model using Roboflow API for automated pothole detection across diverse lighting and weather conditions.

SkolarD (Peer-to-peer Tutoring Application)

May 2025 – June 2025

University of Manitoba

- Built a full-stack Java application with a 3-tier architecture, applying software engineering principles to ensure low coupling, high cohesion, and modular abstraction across components.
- Improved reliability and test coverage through unit, integration, and system testing using Junit and Mockito, along with Git version control and code reviews on GitLab.
- Built modules using Java Swing, SQLite, and design patterns like Factory, Strategy, and Observer.

Relational Movie Database Development

January 2025 – April 2025

University of Manitoba

- Integrated 4M+ movie records from IMDb, TMDb, Oscars, and streaming datasets through data normalization and schema design using Microsoft SQL.
- Developed a Java CLI tool using JDBC for dynamic queries, error handling, and input validation.
- Enhanced data integrity and query efficiency through data cleaning in Python (Pandas, pyodbc).

CurlCUE (Deaf Curling Visual Receiver System)

January 2025 – April 2025

University of Manitoba

- Developed a wireless LED signaling system using C in Arduino IDE on ESP32 microcontrollers to assist Deaf Curling athletes with visual gameplay cues.
- Implemented broadcast-based signal transmission from a central transmitter to multiple receivers, demonstrating distributed communication and synchronization in embedded systems.
- Enhanced portability through battery power management integration and custom 3D-printed enclosures.

Leadership Experience

Director of Operations

Jan 2024 – Current

Engineers Without Borders UM (EWB)

- Led 4 teams comprising 47 members through strategic planning, coordination, and execution of international and local development initiatives focused on sustainability and community impact.
- Delivered two international engineering projects — the Puerto Rico Clean Water Project, which implemented water-filtration systems and food-security initiatives in partnership with the Puerto Rico Food Bank, and the Pakistan Solar Energy Project, providing sustainable electricity to 120+ children in a rural orphanage through a 12 KW hybrid solar installation
- Secured over \$50,000 in funding through community engagement, sponsorship outreach, and internal fundraising efforts, ensuring full financial support for all initiatives.

First Year Engineering Council

January 2020 – April 2020

University Of Manitoba Engineering Society

September 2022 – April 2023

Technical Skills

Programming Languages: Java, Python, C, C#, C++, Verilog, JavaScript, SQL, HTML, MATLAB, DAX, Cypher

Frameworks & Libraries: React, OpenCV, Pandas, JDBC

Parallel & Distributed Computing: CUDA, OpenCL, OpenMP, MPI, Kernel Programming

Databases: PostgreSQL, Microsoft SQL Server, SQLite, MongoDB (NoSQL), Neo4j (Graph)

Cloud & Platforms: Google Cloud Platform (GCP), Microsoft Power Platform (Power BI, Power Apps, Power Automate, SharePoint)

Technologies & Concepts: Blockchain Development, Consensus Algorithms (PoW, PoS), Machine Learning, Deep Learning, Computer Vision, CI/CD Pipelines, Secure Coding